Tensor in ML

Tensor is a data structure. It basically containers stores number.

It is vector and scalar related

- 0D Tensor (scalar):

  (4) , (5)

a array with single value, there will be no dimension.

- 1D Tensor (vector):

  [2, 4, 6, 8, ...] 1-Dimensional array

a list of numbers is a 1D Tensor also called vector, 1D array

but vectors dimension is the no. of elements in that array.
for eg - [1, 2, 3, 4]
4D vector

Rank = 1

Scanned with OKEN Scanner

- 2D Tensor / Matrices

Columns

rows

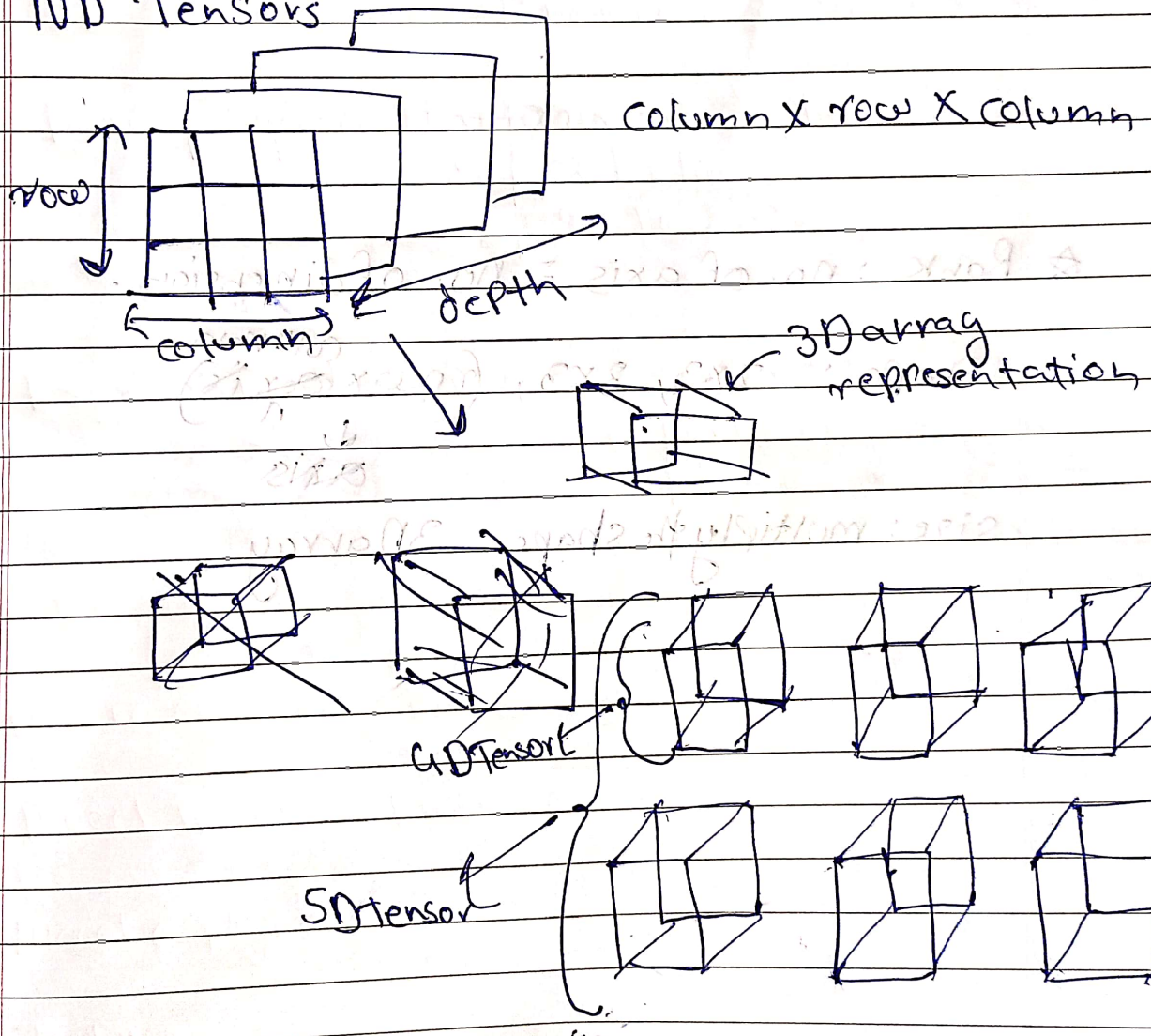
[1, 2, 3]
[4, 5, 6]
[7, 8, 9]

Rank = 2

→ 2D array / Matrices

A 2 dimension Tensor is a 2D Array.

- ND Tensors



Here we see that : 3D array, Tensor is collection of 2D Tensor

4D Tensor is collection of 3D Tensor

5D Tensor is collection of 4D Tensor

⋮

☆ Rank : no. of axis = no. of dimension.

Shape : 2x3, 3x3, (row x column)

axis

size : multiply the shape : 3D array